Alex **Lewandowski**

alex@lewandowski.ca | lewandowskialex.com | AlexLewandowski

I design algorithms so that agents continue to develop their behaviour as they explore the world.

Education

Ph.D. in Computing Science
University of Alberta — 2019 – 2026
» Supervisors: Dale Schuurmans, Marlos C. Machado
» Research topics: continual learning, reinforcement learning, meta-learning

M.Sc. in Statistics
University of Alberta — 2016 – 2018
» Supervisors: Ivor Cribben, Rohana Karunamuni
» Thesis: Recurrent and Bayesian Kernel Learning for Small Data with Applications to Neuroimaging

Honours Bachelor in Mathematics
University of Waterloo — 2012 – 2016
» Major: Mathematical Economics

Papers & Preprints

Universal computation is intrinsic to language model decoding — 01/2026
» **A. Lewandowski**, M. C. Machado, D. Schuurmans
» Preprint

The world is bigger: A computationally-embedded perspective on the big world hypothesis — 12/2025
» **A. Lewandowski**, A. Ramesh, E. Meyer, D. Schuurmans, M. C. Machado
» Neural Information Processing Systems **(NeurIPS spotlight)**

Balancing expressivity and robustness: Constrained rational activations for reinforcement learning — 08/2025
» R. Surdej, M. Bortkiewicz, **A. Lewandowski**, M. Ostaszewski, C. Lyle
» Conference on Lifelong Learning Agents **(CoLLAs oral)**

Plastic learning with deep Fourier features — 04/2025
» **A. Lewandowski**, D. Schuurmans, M. C. Machado
» International Conference on Learning Representations **(ICLR)**

Learning continually by spectral regularization — 04/2025
» **A. Lewandowski**, M. Bortkiewicz, S. Kumar, A. György, D. Schuurmans, M. Ostaszewski, M. C. Machado
» International Conference on Learning Representations **(ICLR)**

The need for a big world simulator: A scientific challenge for continual learning — 08/2024
» S. Kumar, H. J. Jeon, **A. Lewandowski**, B. Van Roy
» Finding The Frame at the Reinforcement Learning Conference **(RLC workshop oral)**

Directions of curvature as an explanation for loss of plasticity — 11/2023
» **A. Lewandowski**, H. Tanaka, D. Schuurmans, M. C. Machado
» Preprint

Reinforcement teaching — 06/2023
» C. Muslimani*, **A. Lewandowski***, D. Schuurmans, M. Taylor, J. Luo
» Transactions on Machine Learning Research **(TMLR)**

ZORB: A derivative-free backpropagation algorithm for neural networks — 12/2020
» V. Ranganathan, **A. Lewandowski**
» Beyond Backpropagation at Neural Information Processing Systems **(NeurIPS workshop oral)**

*Equal contribution

Work Experience

Research Associate (Part-time)
NOAH'S ARK LAB, HUAWEI TECHNOLOGIES CANADA CO., LTD. — 2022 – 2023
» Supervisor: Jun Luo
» Developed a reinforcement learning framework for meta-learning to improve training speed of new agents

Research Associate (Full-time)
NOAH'S ARK LAB, HUAWEI TECHNOLOGIES CANADA CO., LTD. — 2021 – 2022
» Supervisor: Jun Luo
» Designed an evaluation methodology for generalization of reinforcement learning agents, applied to autonomous vehicles

Teaching Assistant
UNIVERSITY OF ALBERTA — 2016 – 2018, 2019 – 2021
» Computing Science: Facilitated discussion groups and developed assignments for a new reinforcement learning course. Provided mentorship and guidance for student projects in foundations of information retrieval
» Mathematical & Statistical Sciences: Offered personalized support to students in all first and second-year classes at the Decima Robinson Support Centre. Conducted help sessions Statistics I/II, Regression Analysis and Time Series Analysis.

Honors & Awards

Alberta Innovates Graduate Student Scholarship
University of Alberta, Department of Computing Science — 2022—2024

President's Doctoral Prize of Distinction
University of Alberta, Department of Computing Science — 2021—2024

NSERC Postgraduate Scholarships – Doctoral
University of Alberta, Department of Computing Science — 2021—2024

Josephine Mitchell Scholarship
University of Alberta, Department of Mathematical and Statistical Sciences — 2017—2018

Profiling Alberta's Graduate Students Award
University of Alberta, Department of Mathematical and Statistical Sciences — 2017

Queen Elizabeth II Graduate Scholarship
University of Alberta, Department of Mathematical and Statistical Sciences — 2016

President's Scholarship
University of Waterloo — 2012

Community

Organizer	RL in Big Worlds at RLC 2026, RLC 2025 (Local Chair), Finding the Frame Workshop at RLC 2025, Openmind Research Institute Banff Retreat 2023-2024, ICLR SSBM Social 2021, NeurIPS SSBM Social 2020, ICLR Reinforcement Learning Social 2020
Reviewer	NeurIPS 2021-2026 (Top Reviewer: 2022, 2025), ICLR 2020-2026 (Outstanding Reviewer: 2021, 2022, 2026), ICML 2021-2024 (Outstanding Reviewer: 2022), RLC 2024-2026, CoLLAs 2025-2026
Other	Academic Director at Computing Science Graduate Student Association 2022—2023